

recognition, handwritten image/writing classification, recurrent neural networks, NLP (natural language processing), word embedding and PDE (partial differential equation).

Keras: This is used for constructing neural networks and machine learning projects. It interoperates with TensorFlow, Theano and R. It runs efficiently on CPUs and GPUs.

It offers standalone modules including optimisers, neural layers, activation functions, initialisation schemes, cost functions and regularisation schemes. It's a simple high-level framework that works on TensorFlow, Theano, etc.

Caffe: This deep learning framework is supported with interfaces like C, C++, Python, MATLAB as well as the command line interface. It provides high speed and transpose capabilities, and it can be used in modelling convolution neural networks (CNN).

It is very important to choose the right model, architecture and open source deep learning framework for industrial solutions. This is a great challenge, though, because of the plethora of options available in the market.

Analyse the business problem and arrive at the solution as the first step. Break the solution into services and understand the models needed for these

services. This will help you narrow down the architecture choices.

The challenge in adopting deep learning technology is the increasing complexity in hardware due to the complex algorithms used. Besides, the standards and protocols are not yet well established and are still evolving.

In addition, there is a tremendous lack of technical expertise in the industry. There are several companies that provide installation, training and support pertaining to deep learning systems, along with online assistance and post-maintenance of deep learning software and the required services. **END** 

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Acknowledgements

The authors would like to thank Kiran M.R. and Raju Alluri of the enterprise architecture group of Wipro Digital for offering their time and support in many ways, while bringing out this article as part of the Architecture Practice efforts.

Disclaimer

The views expressed in this article are those of the authors and Wipro does not necessarily support them.

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